

# Supply-side arm: identifying eta\_s in the BETA closure

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This document is the workplan for the supply-side arm, the cheapest of the candidates that restore variety inside the {BF, ALPHA, BETA} block (section 4.2 of docs/DOCS\_ASSESSMENT.md, ranked candidate 2). House rule: plans normally live in ROADMAP.md or in an ADR; this file exists at your request, and the durable records remain the design file, the registry/scenarios.csv rows, the run manifests and the ADR that authorises the run.jl change. The short answer to the question that prompted it: **yes, under a supply shock BETA separates from ALPHA and from itself** — the row stops being a duplicate and becomes an identification result. The pilot already exists and its numbers are quoted below.

## 1 Why the demand-only matrix cannot see eta\_s

The reported real-GDP metric is the household-consumption Tornqvist index, and under a demand-only shock the price block is pinned by technology alone: the zero-profit conditions  $p_i = c_i(p, w)$  are homogeneous of degree 1 in  $(p, w)$  and contain no demand term, so they determine only the ratio  $p/w = p^*(A)$ . With the CPI numeraire,  $w = 1/\text{CPI}(p^*)$ , and at the calibrated technology ( $A = 1$ , hence  $p^* = 1$ ) that gives  $w = 1$  and  $p = 1$  for **any** demand vector. The BETA supply curve  $L^s = \bar{L} [(w/P)/(w_0/P_0)]^{\eta_s}$  then returns  $\bar{L}$  for every  $\eta_s$ , which is why BETA reproduces ALPHA bit-for-bit (measured: max abs difference in quantities  $1.1 \times 10^{-16}$ , in prices 0).

Two corollaries that the session measured and that bound the design:

- A demand shock of ten times the programme size still gives  $w = 1.0000000000$  and  $\max |p - 1| = 2.66 \times 10^{-15}$ ; no demand-side re-specification can move the real wage.
- The numeraire is not the lever either (DE-0011): re-gauging changes the level of  $(p, w)$ , not their ratio.

The only channel that moves  $w/P$  is technology, i.e. shocks.supply\_shock.

## 2 The pilot that already exists

experiments/probes/probe2\_identification\_and\_shocks.jl, section D: a +20 % productivity shock in sector 1 ( $A_1 = 1.2$ , all other sectors  $A = 1$ ), solved at  $\theta = \epsilon = 0.5$ ,  $\sigma = 0.9$ ,  $\eta = 1$ , warm-started from the stored ALPHA solution. Residual below  $5.4 \times 10^{-13}$  in every cell.

| row                       | $w$ (real wage) | $L$          | real GDP F1  | real GDP F2  | real GDP F3  |
|---------------------------|-----------------|--------------|--------------|--------------|--------------|
| ALPHA<br>( $\eta_s = 0$ ) | 1.0049029745    | 1.0000000000 | 1.0055526180 | 0.9881460294 | 1.0051598783 |
| BETA<br>$\eta_s = 0.5$    | 1.0049029745    | 1.0024484897 | 1.0086846455 | 0.9912768336 | 1.0082906826 |
| BETA $\eta_s = 2$         | 1.0049029745    | 1.0098299882 | 1.0181268156 | 1.0007153159 | 1.0177291649 |
| BETA $\eta_s = 5$         | 1.0049029745    | 1.0247564458 | 1.0372202492 | 1.0198012921 | 1.0368151411 |

Three things follow, and they are the substance of the answer:

1. **BETA moves, ALPHA does not.** Employment is pinned at exactly 1 for ALPHA and rises with  $\eta_s$  for BETA: +0.24 %, +0.98 %, +2.48 % at  $\eta_s = 0.5, 2, 5$ . Under F1 the headline moves from +0.56 % (ALPHA) to +0.87 / +1.81 / +3.72 % — a spread of more than three percentage points across the elasticity grid, against a spread of  $1 \times 10^{-16}$  in the demand-only matrix.

2. **The elasticity is recovered exactly.**  $\ln L / \ln w$  returns 0.500000, 2.000000, 5.000000; the pilot is therefore also a specification test of the implementation.
3. **Prices move at last** ( $\max |p - 1| = 0.196$ ), which is the precondition for the other candidates (markups, capacity) to have anything to bite on.

### 3 Design of the arm

#### 3.1 Shock specification

Three candidate shocks, in ascending cost. The recommendation is to run all three, because they answer different questions.

| id   | shock                                                                                          | what it identifies                                                                                         | note                                                                                                            |
|------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| s1   | $A_1 = 1.2$ , single sector (agriculture)                                                      | the pure identification test                                                                               | already piloted; surgical, but economically arbitrary                                                           |
| prog | $A_i = 1 + \alpha \psi_i$ on the programme's own sectors ( $\alpha \in \{0.05, 0.10, 0.20\}$ ) | the economically relevant case: a green-investment programme bids up the wages of the sectors it buys from | the impulse is 63 % specialised construction; this shock targets exactly the margin the manuscript argues about |
| unif | $A_i = 1.01$ for all $i$                                                                       | the cleanest real-wage experiment; uniform, so no composition contaminates the estimate                    | small magnitude: a uniform 20 % shock would move the real wage by 25 %                                          |

#### 3.2 Cells and run ids

Design `supply_etas` in `experiments/designs/supply_etas.toml`; ids follow `<design>-<labor>-<financing>[-<variant>]` (ADR-0002). One design per shock id keeps the manifests readable: `supply_etas_s1`, `supply_etas_prog`, `supply_etas_unif`.

- Labour rows: ALPHA (the  $\eta_s = 0$  control) and BETA at  $\eta_s \in \{0.5, 1, 2, 5\}$ , e.g. `supply_etas_s1-BETA-F2-et2`.
- Financing: F1, F2, F3 as in the matrix, so the arm nests the published results.
- Shock ladder: three magnitudes per shock id, so the recovered elasticity can be checked for invariance.
- Count:  $5 \times 3 \times 3 = 45$  cells per shock id; the pilot suggests a few seconds per cell, so each design is a single batch.

#### 3.3 What “estimating $\eta_s$ ” means here

Inside the model  $\eta_s$  is a parameter, not an estimate: at a solved cell it is recovered exactly by inversion,

$$\eta_s = \frac{\ln L}{\ln(w/P)},$$

and the pilot returns the input values to six decimals. The arm therefore delivers two publishable quantities rather than a point estimate:

1. **The identification result:** the map  $\eta_s \mapsto (w/P, L, \text{real GDP})$  is strictly monotone and invertible under a supply shock, and degenerate (constant) under a demand-only shock. That is the claim the referee demand R2 actually needs.
2. **The sensitivity band:** the semi-elasticity of the headline to  $\eta_s$ , reported as the spread over  $\eta_s \in [0.5, 5]$  per financing closure — i.e. how much of the published result rests on an elasticity the data do not pin down.

An empirical  $\eta_s$  (from an observed  $\Delta \ln L$  against  $\Delta \ln(w/P)$  pair) then enters the model as a scenario, not as a calibration; the arm is what makes that step informative.

### 3.4 Preregistered signatures

Recorded with the design (ADR-0006), deviations are findings (ADR-0004):

- $\ln L / \ln(w/P)$  reproduces the input  $\eta_s$  to  $1 \times 10^{-6}$  in every BETA cell, and is invariant to the shock magnitude within a shock id.
- ALPHA:  $L = 1$  exactly,  $w > 1$ ; BETA:  $L$  strictly increasing in  $\eta_s$ .
- Monotone ordering of the real-GDP headline in  $\eta_s$  within each financing column.
- Residual gate  $10^{-6}$  (mobile); budget gate  $10^{-9}$ .

### 3.5 Gates and open items to check, not to assume

- **Canary under  $A \neq 1$ .** The ADR-0010 identity  $p \cdot \text{mktc} = S - (I + X - M) + T$  has only ever been asserted at  $p = 1$ . Zero profit still holds, so the derivation should carry, but the terms are now valued at moving prices: assert it on the first supply cells as a diagnostic and record the result (an ADR if it fails).
- **Scale determinacy.** The round-gain criterion  $\max(A_{\text{bill}} / \lambda + (1-m)(1-s)fs) < 1$  uses the baseline intermediate coefficients; a productivity shock changes the optimal input mix at  $\epsilon = 0.5$ , so the criterion must be re-evaluated for the arm.
- **BF in the arm.** The recombination corner ( $\eta = 0$  with  $\eta_s > 0$ ) becomes informative only here, but it inherits the identification defect of ADR-0017: report its aggregates, never its quantities, until sector-specific wages land.

## 4 Workplan

1. **ADR-0018 (blocking).** `experiments/run.jl` hard-codes `Shocks(ones(N), ones(N), zeros(N))` in `build_cell_model` and `solve_cell`; the arm needs a per-cell supply-shock source and magnitude from the design file. The ADR fixes the schema, the scenario-id convention for  $\eta_s$  variants, and the rule that the baseline reference stays the no-shock one (real GDP remains “against the baseline”).
2. **Bridge, before the ADR lands.** Extend the pilot to the full grid as `experiments/probes/probe5_supply_arm.jl` (read-only, no `src/` change), so the numbers exist and the signatures can be preregistered against measurement rather than expectation.
3. **Registry.** Append planned rows to `registry/scenarios.csv` for every cell (45 per shock id), with the parameter cells, the shock id and magnitude, and `eta_s`.

4. **Design and preregistration.** Write `experiments/designs/supply_etas_<id>.toml`, then

```
julia --project=. experiments/run.jl --list supply_etas_s1
julia --project=. experiments/run.jl --preregister supply_etas_s1 --actor hermes-agent
```

and commit the preregistration record before any cell is created.

5. **Execution.**

```
julia --project=. experiments/run.jl --design supply_etas_s1 --actor hermes-agent
```

then `supply_etas_prog` and `supply_etas_unif`.

6. **Gates after every batch.**

```
julia --project=. scripts/status.jl          # 0 warnings
julia --project=. scripts/check_repo.jl      # 0 violations
julia --project=. -e 'using Pkg; Pkg.test()'
```

7. **Write-up.** A paper table citing the `supply_etas-*` run ids: rows ALPHA and BETA ( $\eta_s$  grid), columns F1/F2/F3, cells = real GDP and employment, plus the recovered elasticity column. Update `docs/DOCS_ASSESSMENT.md` (section 4.2, the ranked proposal) and `registry/closures.toml` [`labor.BETA`] open gates; one `docs/log/2026-09.md` entry.

## 5 What the arm does not do

The arm moves prices **through technology only**: the zero-profit block contains no demand term, so  $p \neq 1$  requires  $A \neq 1$ . Put the demand-only programme back in and the matrix returns to  $p = 1$ ,  $w = 1$ ,  $L = \bar{L}$  – BETA collapses

onto ALPHA again. The supply-side arm is therefore a **different experiment**, not a repair of the published 5 x 3 matrix: it identifies  $\eta_s$  and it prices the elasticity, but it does not restore variety inside the demand-shock block.

Variety in the matrix the paper publishes requires the price block to depend on demand, i.e. one of:

- **demand-sensitive factor prices** — sector-specific wages (ranked candidate 1): N sectoral labour markets make  $w_i$ , and through zero profit  $p_i$ , functions of sectoral labour demand. Zero profit is preserved (no rents to dispose of), it repairs the ADR-0017 defect, and it is the only variant that is *earned* by the model rather than assumed: the new parameters are sectoral supply elasticities of the same family as  $\eta_s$ .
- **a demand-sensitive markup**  $\mu_i = 1 + \kappa (y_i/y_{i0} - 1)$ : nested at  $\kappa = 0$  (the current model), a few lines of code, and a clean sensitivity ladder. But markup income is profit, so it needs a profit-income-and-spending closure (who receives the rent, and how it is spent) or the accounting identity behind the canary breaks — that is the real cost, not the code.
- **a fixed factor with capacity**: the standard short-run mechanism, but the largest surgery — a second factor, a rental schedule and a rental-income closure.

Once prices respond to demand, BETA separates from ALPHA in the demand matrix as well: the numeraire then delivers a real wage that moves with the shock, so  $L^s = \bar{L} (w/P)^{\eta_s}$  bites. The sequencing this implies: supply-side arm first (cheap, earns the identification claim for  $\eta_s$ ), then sector-specific wages (the repair that gives the published matrix variety), with the markup ladder kept as a robustness arm rather than as the headline mechanism.

## 6 Risks

- **Solver.** The pilot used `solve_beta` with eight continuation rungs and passed at  $\eta_s = 5$ ; the prog shock at  $\alpha = 0.20$  hits the construction block harder, so budget more rungs and keep the warm start.
- **Magnitude.** A uniform shock must stay small (one to five per cent); the single-sector shock can be large because its consumption weight is small.
- **Interpretation.** The arm changes the *experiment*, not the closure; nothing in `src/` moves. It does not fix the BF row (that is candidate 1, sector-specific wages), and it does not make  $\eta_s$  identified without a supply shock — the demand-only matrix stays a null result for this parameter, which is itself the finding.

## 7 Deliverables

- `experiments/designs/supply_etas_{s1,prog,unif}.toml` with preregistration records.
- One run manifest per cell under `runs/supply_etas-*/`, rows in `registry/scenarios.csv`, index rows in `runs/index.csv`.
- ADR-0018 (design schema for supply shocks) and, if the canary or the round-gain check fails under  $A \neq 1$ , a follow-up ADR.
- `paper/tables/supply_etas_flows.md`, generated from the manifests and citing the run ids (ADR-0004).